

Introduction

- **Activation steering** (or “representation engineering”) is a powerful approach to influence model behavior **at inference time** by **directly modifying activations** during generation.
- Most existing methods are **open-loop**, resulting in suboptimal interventions that can degrade model performance.
- We pose activation steering as a **control synthesis** problem, leveraging **optimal control theory** to achieve robust model control.

Problem Setting & Notation

- Assume the **linear representation hypothesis**: semantic concepts are modulated by a **single direction** in activation space.
- Define an **activation-steered LLM** $M_{\text{steered}} : \mathcal{V}^k \rightarrow \mathcal{V}$

$$M_{\text{steered}}(q) := \text{Unembed} \circ \rho_\ell \circ \dots \circ \rho_1 \circ \text{Embed}(q), \quad (1)$$

with **controlled transformer layers**, where z_k and u_k are the residual stream vector and steering intervention at layer k :

$$z_{k+1} := \rho_k(z_k, u_k) := \phi_k(z_k) + u_k. \quad (2)$$

- **Objective**: construct $\{u_k\}_{k=l}^{\ell-1}$ so that M_{steered} achieves a desired behavior (e.g., toxicity reduction).

Local Linearity in LLMs

We empirically assess the linear-time-varying (LTV) model of LLM dynamics by measuring the **alignment between dominant modes** of layer-wise Jacobians $\frac{\partial \phi_k}{\partial z}$ along different trajectories.

$$\text{sim}_m(W, X) := \frac{\|A\|_*}{\|\Sigma_W^{(m)}\|_F \|\Sigma_X^{(m)}\|_F} \quad (3)$$

$$A := (U_W^{(m)} \Sigma_W^{(m)})^\top U_X^{(m)} \Sigma_X^{(m)},$$

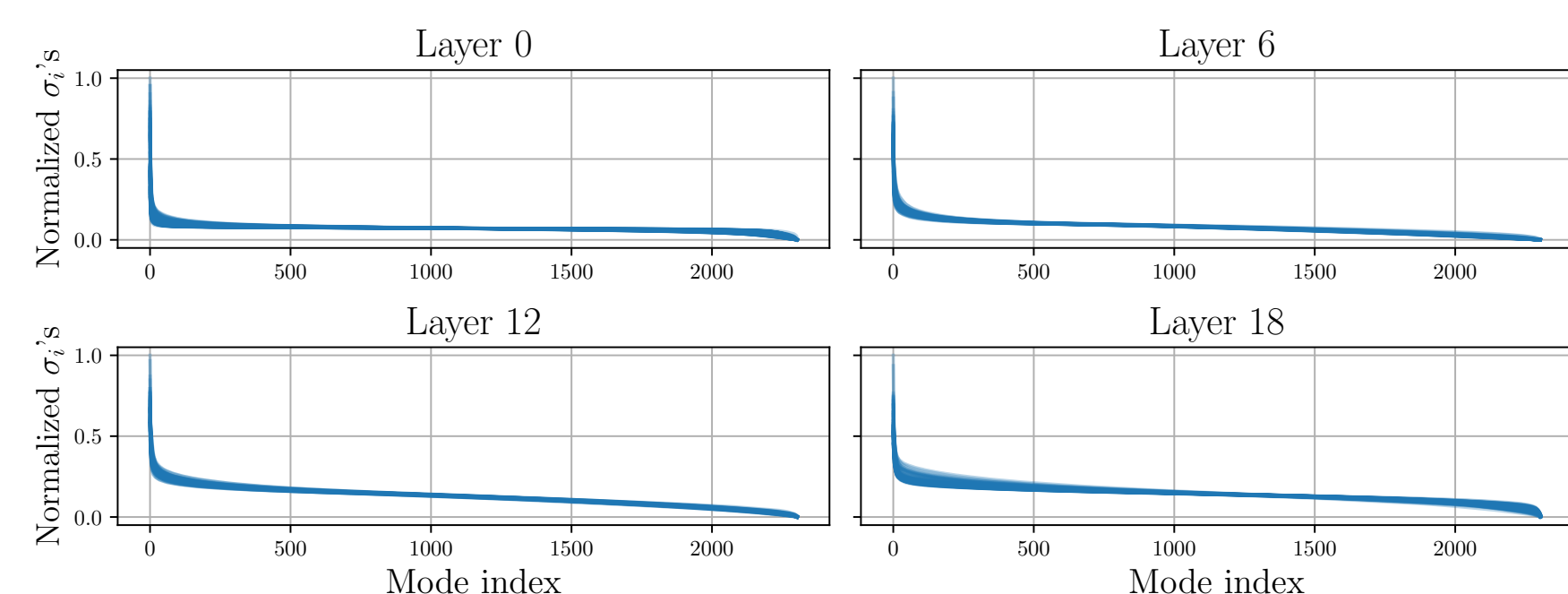


Figure 1. Range of normalized singular values across layers of Gemma-2-2B for Jacobians corresponding to random nominal trajectories. We observe that the spectra are dominated by a few modes.

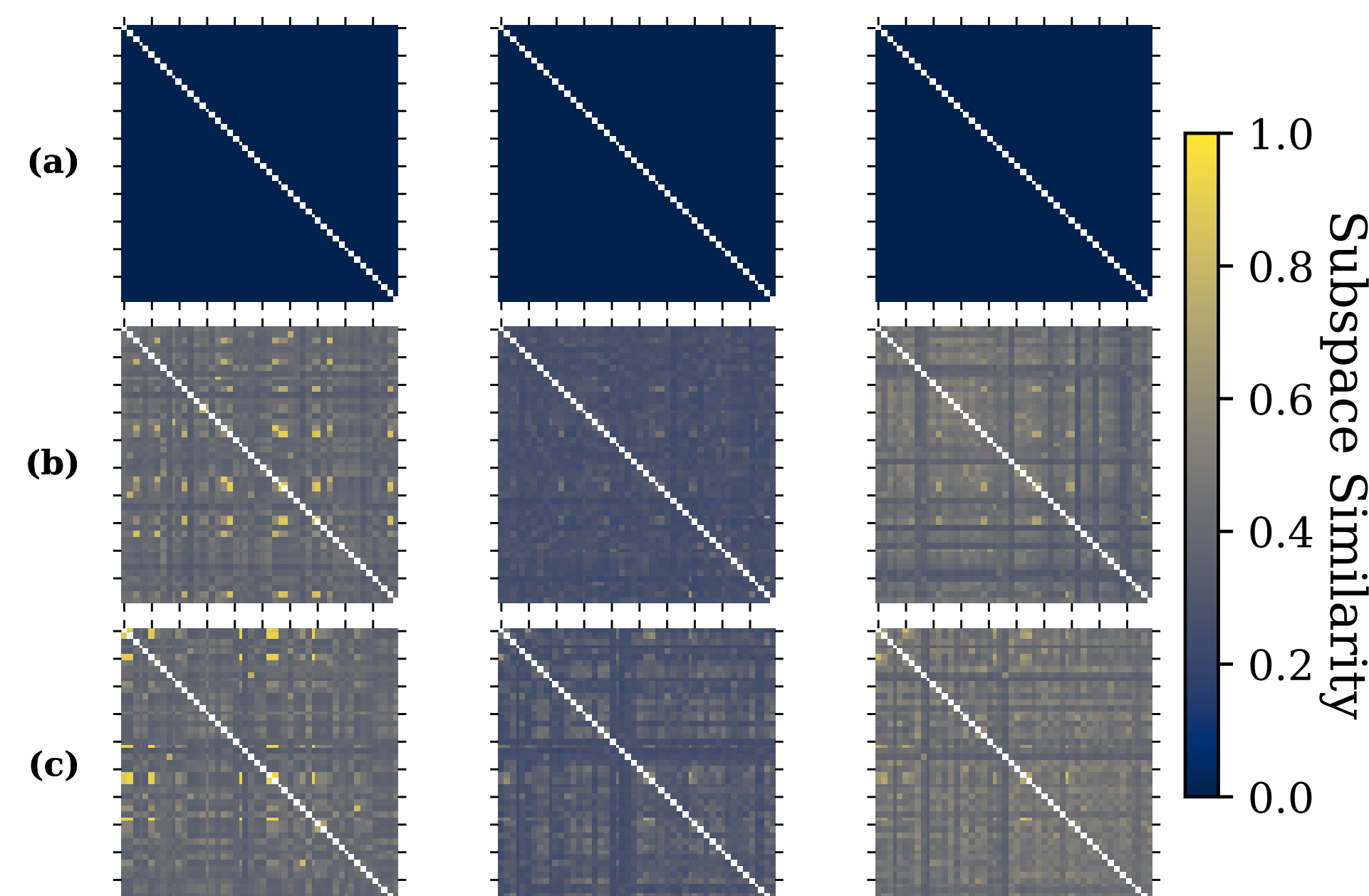


Figure 2. Initial, intermediate, and final linearized-layer alignment for Gemma-2-2B along top dynamical modes. Each plot contains comparisons across 50 matrices (one matrix per row/column), with lighter pixels corresponding to stronger alignment. (a) randomly sampled matrices, (b) randomly sampled nominal Jacobians of differing concepts, and (c) Jacobians corresponding the prompts related to the concept “Cloud.”

Key Takeaway

LLM transformer blocks are locally linear $\implies$ classical linear control yields a steering strategy that

- is *training-free* and *closed-form*
- *adapts online* to reject stochasticity
- *minimizes invasiveness*
- achieves *SOTA alignment* with minimal fluency loss.

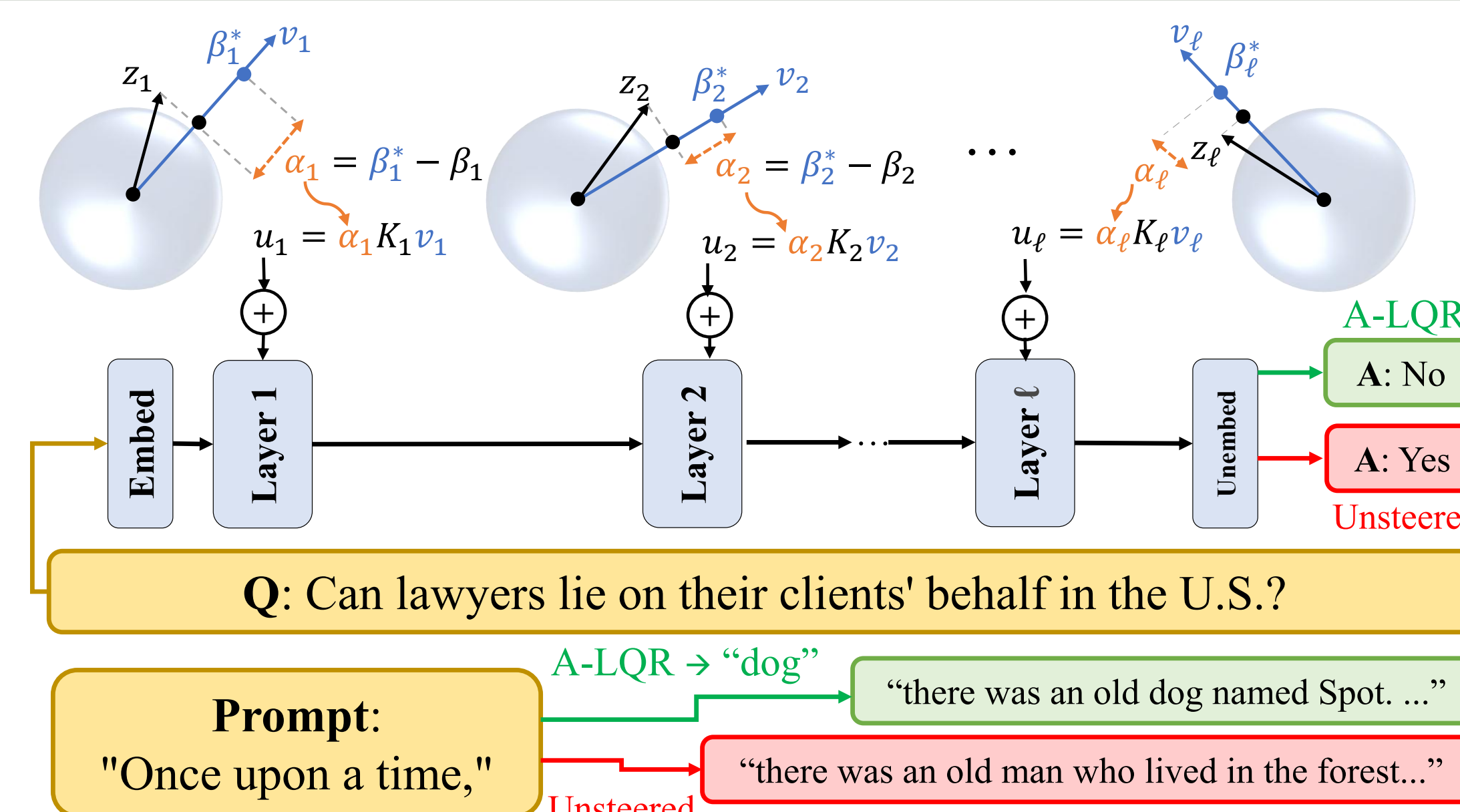


Figure 3. **Overview.** Our method, A-LQR, computes a steering intervention u_k that minimizes the deviation between the semantic feature value β_k at the current activation z_k and a desired target β_k^* .

Step 1: Semantic Steering Objective

- Given contrastive prompt datasets with and without a target concept, e.g., $\mathcal{D}_+ = \{\text{Toxic Prompts}\}$ and $\mathcal{D}_- = \{\text{Nontoxic Prompts}\}$, we compute the mean-difference **contrastive vector** e_k :

$$z_{k,+} := \frac{1}{|\mathcal{D}_+|} \sum_{i=1}^{N_+} z_{k,+}^{(i)}, \quad z_{k,-} := \frac{1}{|\mathcal{D}_-|} \sum_{i=1}^{N_-} z_{k,-}^{(i)},$$

$$e_k := z_{k,+} - z_{k,-}, \quad \mu_k := \|e_k\|_2, \quad v_k := e_k / \mu_k$$

- Define the **feature strength** of a realized activation as $\beta_k := v_k^\top z_k$.
- **Linear Feature Setpoint (LFS)**: Layer-adaptive feature strength target

$$\beta_k^* := \lambda \mu_k,$$

where λ is a user parameter. From the LFS we derive a per-layer **online feedback signal**:

$$\delta z_k := -\alpha_k v_k, \quad \alpha_k := \beta_k^* - \beta_k.$$

Step 2: Tracking Semantic Setpoints

- **Linearize** transformer blocks ϕ about a nominal trajectory, to yield the **LTV dynamics**:

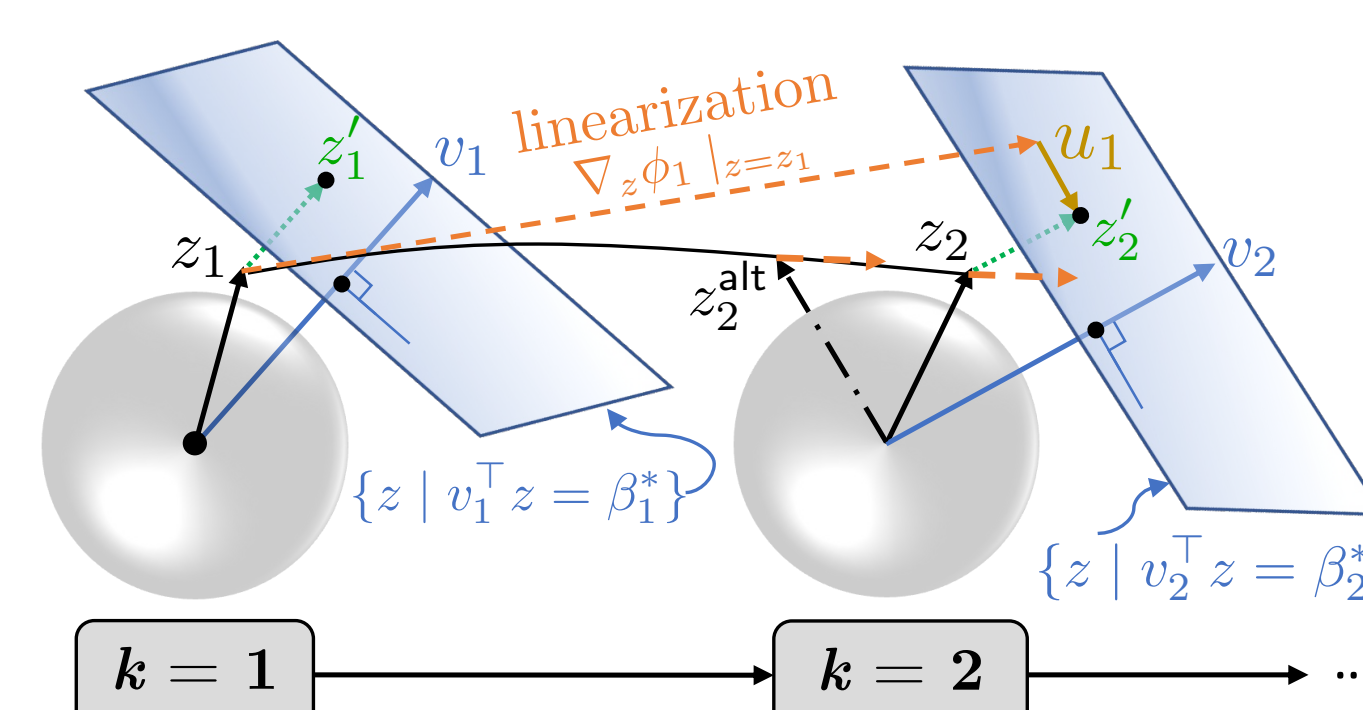
$$\delta z_{k+1} \approx A \delta z_k + u_k, \quad A := \frac{\partial \phi_k}{\partial z} \Big|_{\bar{z}}$$

- Solve the LQR tracking problem with **Ricatti recursions** to get a set of gains $\{K_k\}_k$.
- **Activation LQR (A-LQR)**: At each layer, A-LQR tracks LFS via the control perturbation:

$$u_k^* = (\beta_k - v_k^\top z_k) \cdot K_k v_k$$

Key Properties:

- **Training free**: gains are precomputed offline.
- **Reusable gains**: Once computed, $\{K_k\}_k$ can be fixed across prompts.



Theoretical Analysis of A-LQR

Approximating the LLM as an LTV system enables LQR tools but is not exact – we derive a **theoretical bound on A-LQR tracking error**.

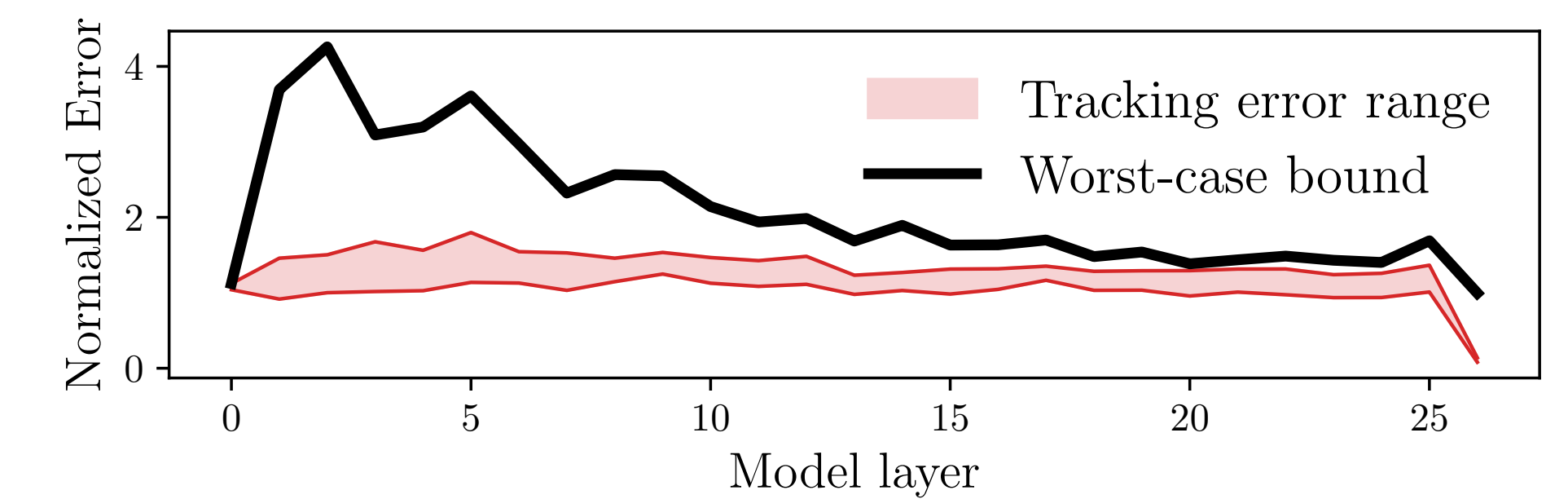


Figure 4. Empirical tracking error satisfies the theoretical bound.

Results: Concept Steering

- The model is prompted with “**Once upon a time**” – then A-LQR is used to induce various concepts.
- We observe that scaling the LFS parameter λ achieves **fine-grained control** of each concept.

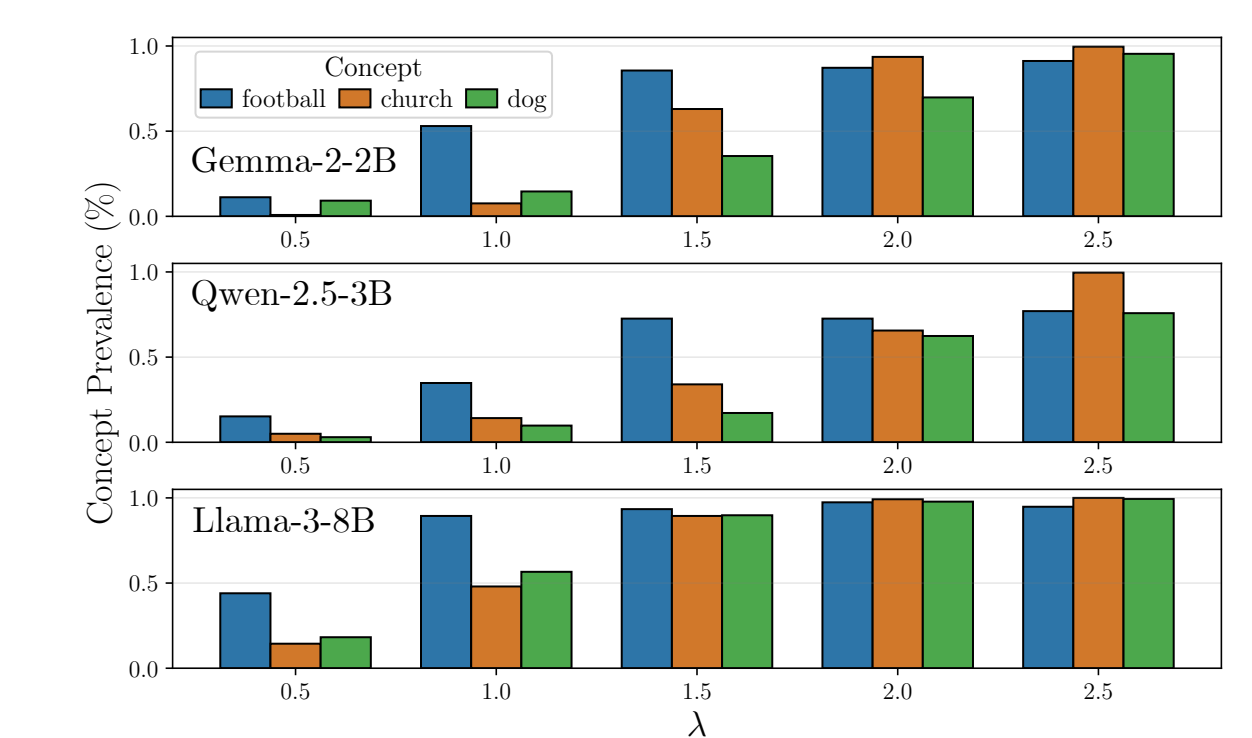


Figure 5. Concept prevalence in LLM generations against different setpoint targets.

Results: Toxicity & Truthfulness

The model is prompted with various offensive or misleading prompts. A-LQR mitigates harmful or misleading responses.

- **Closed-loop** control achieves stronger steering results than open-loop baselines.
- **Minimally invasive** interventions preserve model fluency through explicit control penalty.

Table 1. Toxicity mitigation results on Gemma-2-2B.

Method	CLS Tox. (%) (↓)	MMLU (%) (↑)	PPL (↓)
Original	4.16 ± 0.54	54.54 ± 1.71	8.95 ± 0.07
2nd Best Baseline	0.58 ± 0.10	52.52 ± 2.07	11.62 ± 0.06
Best Baseline	0.50 ± 0.24	54.62 ± 1.22	8.70 ± 0.06
A-LQR	0.18 ± 0.08	53.56 ± 1.62	12.26 ± 0.08

Table 2. Truthfulness results on Qwen2.5-14B.

Method	T-I (↑)	True (%) (↑)	Info (%) (↑)	MMLU (%) (↑)
Original	53.17 ± 0.52	55.15 ± 0.52	96.40 ± 0.22	78.80 ± 0.00
2nd Best Baseline	57.01 ± 0.59	59.09 ± 0.88	96.47 ± 0.55	79.04 ± 0.08
Best Baseline	71.56 ± 1.26	75.03 ± 1.12	95.37 ± 0.67	78.08 ± 0.77
A-LQR	76.28 ± 0.87	80.81 ± 0.60	94.39 ± 0.54	71.66 ± 1.48

Results: Jailbreaking

Table 3. Jailbreaking results on Gemma-2-9b Instruct

Method	ASR (↑)	Refusal % (↓)
Original	0	1.0
Best baseline	0.13	0.68
A-LQR	0.14	0
A-LQR+	0.77	0

A-LQR can enforce compliance with prompts the model typically would refuse.

- **A-LQR+** intervenes on all token positions.

Conclusion

- We study **local linearity** in LLMs and use it to efficiently design **feedback controllers** for activation steering.
- We propose **A-LQR**, an extension of LQR to LLM dynamics, which tracks **LFS**, a novel, semantically-meaningful feedback signal.
- A-LQR tracking LFS achieves **state-of-the-art results** across activation steering tasks, while **preserving model fluency**.